2026 AMC 12B

Solutions

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1 D	2 C	3 B	4 C	5 C	6 B	7 D	8 B	9 B	10 B	11 B	12 D	13 D
14 B	15 A	16 E	17 C	18 E	19 C	20 D	21 D	22 A	23 C	24 A	25 D	

Problem 1

A souvenir shop sells solid metal sculptures of Achilles in two sizes. One is 6 cm tall and one is 8 cm tall. Proportions between the two sculptures are preserved. The smaller sculpture has a mass of 486 grams. What is the mass, in grams, of the larger sculpture?

A 648 **B** 864 **C** 972 **D** 1152 **E** 1458

Solution. Mass is proportional to volume, and for similar solids volume scales with the cube of length. The height ratio is $\frac{8}{6} = \frac{4}{3}$, so

$$486 \cdot \left(\frac{4}{3}\right)^3 = 486 \cdot \frac{64}{27} = 1152.$$

Answer: **D**

Problem 2

Bees occupy the 8 corners of a $1 \times 2 \times 3$ rectangular box. The bees can fly inside the box, but cannot fly farther than $\frac{1}{2}$ unit from their respective corners. What is the volume, in cubic units, of the region inside the box that cannot be reached by any bee?

A $6 - \frac{\pi}{3}$ **B** $6 - \frac{\pi}{4}$ **C** $6 - \frac{\pi}{6}$ **D** $6 - \frac{\pi}{8}$ **E** $6 - \frac{\pi}{12}$

Solution. Each bee can reach an eighth of a ball of radius $\frac{1}{2}$ around its corner, and since every edge of the box has length at least 1, no two of these eighths overlap.

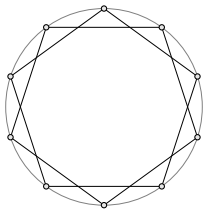
Eight eighths make one whole ball, of volume $\frac{4}{3}\pi \cdot \frac{1}{8} = \frac{\pi}{6}$. So the unreachable volume is $6 - \frac{\pi}{6}$.

Answer: **C**

Problem 3

There are 2026 equally spaced points that all lie on a circle. A line segment is drawn between two points if there is exactly one point between them along the circle. Into how many regions will this divide the interior of the circle? For example, with 10 equally spaced points there are 21 interior regions.

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A 4052 **B** 4053 **C** 4054 **D** 6079 **E** 6081

Solution. Treat the figure as a planar graph and use Euler's formula $V - E + F = 2$.

Segment P_iP_{i+2} crosses only $P_{i-1}P_{i+1}$ and $P_{i+1}P_{i+3}$, so there are 2026 crossing points, and

$$V = 2026 + 2026 = 4052.$$

Each segment is cut into 3 pieces by its two crossings, and the circle is cut into 2026 arcs, so $E = 3 \cdot 2026 + 2026 = 8104$.

Then $F = 2 - 4052 + 8104 = 4054$, and removing the outside face leaves 4053 regions. (With 10 points the same count gives 21, matching the example.)

Answer: **B**

Problem 4

For how many integers n is $\dfrac{n^2-1}{n+2}$ also an integer?

A 2 **B** 3 **C** 4 **D** 6 **E** infinitely many

Solution. Polynomial division gives

$$\frac{n^2-1}{n+2} = n - 2 + \frac{3}{n+2},$$

so we need $n+2$ to divide 3. That means $n+2 \in \{-3, -1, 1, 3\}$, so $n \in \{-5, -3, -1, 1\}$, which is 4 integers.

Answer: **C**

Problem 5

Let $a = \log_2 27$, let $b = \log_3 25$, and let $c = \log_{32} 5$. What is $\dfrac{ab}{c}$?

A 25 **B** 27 **C** 30 **D** 32 **E** 36

Solution. Switch everything to natural logs:

$$\frac{ab}{c} = \frac{\ln 27}{\ln 2} \cdot \frac{\ln 25}{\ln 3} \cdot \frac{\ln 32}{\ln 5} = \frac{3\ln 3}{\ln 2} \cdot \frac{2\ln 5}{\ln 3} \cdot \frac{5\ln 2}{\ln 5} = 30.$$

Answer: **C**

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Problem 6

The integer n satisfies the inequalities

$$\frac{n}{100} < \frac{1}{5 \cdot 10} + \frac{1}{10 \cdot 15} + \frac{1}{15 \cdot 20} + \cdots + \frac{1}{90 \cdot 95} + \frac{1}{95 \cdot 100} < \frac{n+1}{100}.$$

What is n?

A 2 **B** 3 **C** 4 **D** 5 **E** 6

Solution. Each term is $\frac{1}{25k(k+1)} = \frac{1}{25}\left(\frac{1}{k} - \frac{1}{k+1}\right)$ for $k = 1, \ldots, 19$, so the sum telescopes:

$$\frac{1}{25}\left(1 - \frac{1}{20}\right) = \frac{19}{500} = \frac{3.8}{100}.$$

So $n < 3.8 < n + 1$, giving $n = 3$.

Answer: **B**

Problem 7

Sharon tosses a fair coin six times and keeps track of how many heads and tails she has gotten. Her first toss is heads. What is the probability that after every toss Sharon's count of heads differs from her count of tails?

A $\frac{1}{8}$ **B** $\frac{3}{16}$ **C** $\frac{1}{4}$ **D** $\frac{5}{16}$ **E** $\frac{3}{8}$

Solution. Track $d = \#H - \#T$. It starts at $d = 1$, moves by ± 1 each toss, and we need it never to hit 0 in the remaining five tosses.

Count the bad paths with the reflection trick. Reflecting a path from 1 up to the first time it hits 0 gives a path from -1, so the bad paths that finish above 0 match the paths from -1 that finish above 0. Every path from 1 that finishes at or below 0 is bad, so

$$\#\text{good} = \#\{\text{paths from } 1 \text{ ending} > 0\} - \#\{\text{paths from } -1 \text{ ending} > 0\}.$$

A path from 1 with u up-steps ends at $2u - 4$, so it ends above 0 when $u \geq 3$, which happens in $10 + 5 + 1 = 16$ ways. A path from -1 ends at $2u - 6$, which is positive when $u \geq 4$, in $5 + 1 = 6$ ways.

So there are $16 - 6 = 10$ good paths out of 32, and the probability is $\frac{10}{32} = \frac{5}{16}$.

Answer: **D**

Problem 8

A regular octahedron has 8 equilateral triangle faces with edges of length $\sqrt{2}$. A plane is perpendicular to the line segment joining two opposite vertices of the octahedron and intersects that line segment $\frac{1}{2}$ unit from one of its endpoints. What is the area of the intersection of that plane and the octahedron?

A $\frac{1}{4}$ **B** $\frac{1}{2}$ **C** $\frac{1}{2}\sqrt{2}$ **D** $\frac{3}{4}$ **E** $\sqrt{2}$

Solution. Edges of length $\sqrt{2}$ are exactly what you get from the vertices $(\pm 1, 0, 0)$, $(0, \pm 1, 0)$, $(0, 0, \pm 1)$. So the octahedron is

$$|x| + |y| + |z| \leq 1,$$

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and the segment joining $(0, 0, \pm 1)$ has length 2. The plane is $z = \frac{1}{2}$, and the cross-section is $|x| + |y| \leq \frac{1}{2}$. That is a square with diagonals of length 1, so its area is $\frac{1}{2} \cdot 1 \cdot 1 = \frac{1}{2}$.

Answer: **B**

Problem 9

Suppose that a and b are positive integers satisfying $20a + 26b = 2026$. What is the least possible value of $a + b$?

A 71 **B** 80 **C** 83 **D** 86 **E** 88

Solution. Halving, $10a + 13b = 1013$, so $b = \frac{1013-10a}{13}$ and

$$a + b = \frac{1013 + 3a}{13}.$$

This is smallest when a is smallest. We need $13 \mid 1013 - 10a$. As $1013 \equiv 12 \equiv -1 \pmod{13}$ and $10^{-1} \equiv 4 \pmod{13}$, this means $a \equiv -4 \equiv 9 \pmod{13}$.

So $a = 9$, $b = 71$, and $a + b = 80$.

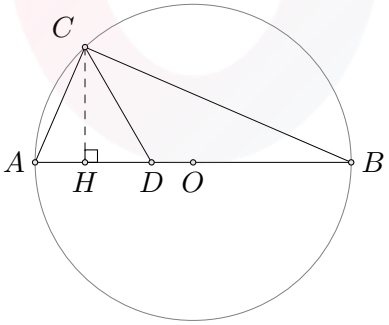
Answer: **B**

Problem 10

Point C lies on a circle with diameter $\overline{AB}$. Point D lies on $\overline{AB}$, 14 units from A, 24 units from B, and 16 units from C. What is the area, in square units, of $\triangle ABC$?

A $186\sqrt{2}$ **B** $152\sqrt{3}$ **C** 264 **D** $187\sqrt{2}$ **E** $108\sqrt{6}$

Solution.



Since AB is a diameter, $\angle ACB = 90°$, so $CA^2 + CB^2 = 38^2 = 1444$. *Stewart's Theorem* for the cevian CD gives

$$24\,CA^2 + 14\,CB^2 = 38\left(16^2 + 14 \cdot 24\right) = 22496.$$

Substituting $CB^2 = 1444 - CA^2$ gives $10\,CA^2 = 2280$, so $CA^2 = 228$ and $CB^2 = 1216$. Then

$$[ABC] = \tfrac{1}{2}\,CA \cdot CB = \tfrac{1}{2}\sqrt{228 \cdot 1216} = \tfrac{1}{2} \cdot 304\sqrt{3} = 152\sqrt{3}.$$

Answer: **B**

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Problem 11

A list contains 2026 statements, each of which is either true or false. The first statement is "The number of false statements in this list is at least 1001." The second statement is "The number of false statements in this list is at least 1002." In general, the kth statement in the list is "The number of false statements in this list is at least $k + 1000$." How many statements in the list are true?

A 342 **B** 513 **C** 1013 **D** 1026 **E** 1513

Solution. Say F statements are false. Statement k is true exactly when $F \geq k + 1000$, i.e. when $k \leq F - 1000$. So the true statements are the first $F - 1000$ of them, and

$$(F - 1000) + F = 2026 \implies F = 1513.$$

That leaves $2026 - 1513 = 513$ true statements.

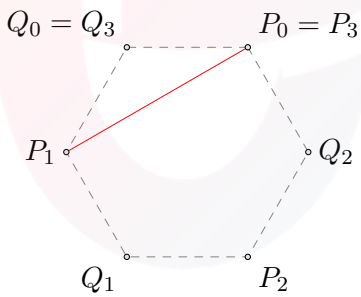
Answer: B

Problem 12

Polly and Quentin are dancing. They start 1 meter apart. Polly rotates 120° clockwise around Quentin while he remains still, staying 1 meter away throughout. Then Quentin rotates 120° clockwise around Polly while she remains still, staying 1 meter away throughout. They repeatedly do this pair of moves from their new positions. How many meters from where she started is Polly after they have each moved 10 times?

A 0 **B** $\frac{\sqrt{3}}{2}$ **C** 1 **D** $\sqrt{3}$ **E** 2

Solution.



A pair of moves is a 120° clockwise rotation about Quentin followed by a 120° clockwise rotation about Polly's new position. Two rotations compose to a rotation by the sum of the angles, here 240° clockwise, about some fixed point. So each pair of moves rotates the whole couple by 240° about the same centre, and after 3 pairs everyone is back home.

As $10 = 3 \cdot 3 + 1$, after ten pairs Polly is where she was after one pair. In that first pair she swung 120° around Quentin at distance 1 and then stayed still, so she ends up a chord of a unit circle subtending 120° away from her start, which has length $2 \sin 60° = \sqrt{3}$. (The figure shows the three positions of each dancer.)

Answer: D

Problem 13

A jar contains 12 green marbles and 8 red marbles. In a game, Anika and Ben take turns drawing a random marble from the jar, with replacement. If Ben ever draws a red marble, he wins, and if

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Anika ever draws a green marble, she wins. Otherwise, the game continues with the next player's turn. The probability that Ben eventually wins is A if Anika goes first and B if Ben goes first. What is $B - A$?

A $\frac{2}{25}$ **B** $\frac{4}{25}$ **C** $\frac{4}{19}$ **D** $\frac{6}{19}$ **E** $\frac{9}{20}$

Solution. Each draw is red with probability $\frac{2}{5}$ and green with probability $\frac{3}{5}$.

Suppose Ben goes first. He wins at once with probability $\frac{2}{5}$. Otherwise Anika draws, and with probability $\frac{2}{5}$ she misses too, which puts us back at the start. So

$$B = \frac{2}{5} + \frac{3}{5} \cdot \frac{2}{5} B \implies \frac{19}{25} B = \frac{2}{5} \implies B = \frac{10}{19}.$$

If Anika goes first, Ben can only win if she misses first, which happens with probability $\frac{2}{5}$, and then it is Ben's turn. So $A = \frac{2}{5} B = \frac{4}{19}$, and

$$B - A = \frac{6}{19}.$$

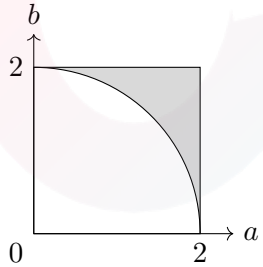
Answer: **D**

Problem 14

A line segment is drawn between two points on the graph of $y = x^2$ whose x-coordinates have been chosen independently and uniformly at random in the interval $[0, 2]$. The probability that the y-coordinate of the midpoint of the segment is greater than 2 lies in which of the following intervals?

A $(0.20, 0.21)$ **B** $(0.21, 0.22)$ **C** $(0.22, 0.23)$ **D** $(0.23, 0.24)$ **E** $(0.24, 0.25)$

Solution.



With endpoints (a, a^2) and (b, b^2), the midpoint has height $\frac{a^2+b^2}{2}$. So we want $a^2 + b^2 > 4$ for (a, b) uniform in $[0, 2]^2$.

That is the part of the square outside the quarter disc of radius 2, shaded in the figure, with area $4 - \pi$. The probability is

$$\frac{4 - \pi}{4} = 1 - \frac{\pi}{4} \approx 0.2146.$$

Answer: **B**

Problem 15

What is the value of

$$\log_2\left(\sin\frac{\pi}{8}\right) + \log_2\left(\sin\frac{3\pi}{8}\right) + \log_2\left(\sin\frac{5\pi}{8}\right) + \log_2\left(\sin\frac{7\pi}{8}\right)?$$

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A -3 **B** $-2\sqrt{2}$ **C** -2 **D** $-\frac{3}{2}$ **E** $-\sqrt{2}$

Solution. Since $\sin\frac{5\pi}{8} = \sin\frac{3\pi}{8}$ and $\sin\frac{7\pi}{8} = \sin\frac{\pi}{8}$, the product of the four sines is $\left(\sin\frac{\pi}{8}\sin\frac{3\pi}{8}\right)^2$. By product to sum,

$$\sin\tfrac{\pi}{8}\sin\tfrac{3\pi}{8} = \tfrac{1}{2}\left(\cos\tfrac{\pi}{4} - \cos\tfrac{\pi}{2}\right) = \tfrac{\sqrt{2}}{4},$$

so the product is $\frac{1}{8}$ and the sum of logs is $\log_2\frac{1}{8} = -3$.

Answer: **A**

Problem 16

Annie, Bobby, and Casey each receive an integer, not necessarily distinct, from 1 through 10. In how many ways can the numbers be given so that their sum is divisible by 7?

A 139 **B** 140 **C** 142 **D** 143 **E** 145

Solution. Pick Annie's and Bobby's numbers first, in 100 ways. Casey then needs a residue fixed mod 7. Among $1, \ldots, 10$, the residues $1, 2, 3$ appear twice and the rest once, so Casey has 2 choices if $a + b \equiv 4, 5, 6 \pmod 7$, and 1 otherwise. So the answer is

$$100 + \#\{(a, b) : a + b \equiv 4, 5, 6 \pmod 7\}.$$

Let c_r count the numbers $1, \ldots, 10$ with residue r, so $c_1 = c_2 = c_3 = 2$ and $c_0 = c_4 = c_5 = c_6 = 1$. Summing $c_r c_{s-r}$ over r gives 16 pairs with $a + b \equiv 4$, 15 with $a + b \equiv 5$ and 14 with $a + b \equiv 6$.

The total is $100 + 16 + 15 + 14 = 145$.

Answer: **E**

Problem 17

Belinda follows a path on the plane that is entirely contained within the square region $0 \le x, y \le 4$. Call her path *dignified* if and only if each step is one unit up, down, or to the right, the path does not revisit any vertex, and the path starts at $(a, 0)$ and ends at $(b, 0)$, where a and b are integers with $a < b$, and touches the x-axis only at these two points. How many dignified paths can Belinda follow?

A 174 **B** 340 **C** 448 **D** 512 **E** 970

Solution. Since Belinda never steps left and never revisits a vertex, in any column she only moves in one vertical direction. So a dignified path is fixed by the heights of its rightward steps.

Going from $(a, 0)$ to $(b, 0)$ she takes $b - a$ steps to the right, each at a height in $\{1, 2, 3, 4\}$, since height 0 is off limits. Any such choice of heights works, so there are 4^{b-a} paths.

For $0 \le a < b \le 4$, a gap of $b - a = d$ occurs $5 - d$ times, so the total is

$$4 \cdot 4 + 3 \cdot 16 + 2 \cdot 64 + 1 \cdot 256 = 448.$$

Answer: **C**

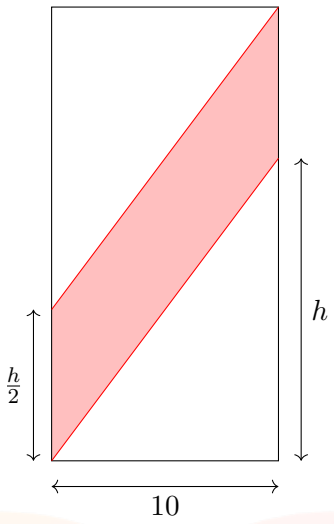
Problem 18

A ribbon wraps around a pole in such a way that equal amounts of area are covered and not covered by the ribbon. The width of the ribbon is 4 cm, and the circumference of the pole is 10 cm. In centimeters, how much of the pole height is needed for the ribbon to wrap around exactly once?

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A $\frac{15}{2}$ **B** 8 **C** 10 **D** $\frac{25}{2}$ **E** $\frac{40}{3}$

Solution.



Unroll the pole, and let h be the height used by one full wrap. Over one wrap the pole surface has area $10h$. The ribbon's edge runs along the hypotenuse of a 10 by h right triangle, so the ribbon used is a strip of width 4 and length $\sqrt{100+h^2}$.

Half the surface is covered, so

$$4\sqrt{100+h^2} = \tfrac{1}{2}\cdot 10h \implies 16(100+h^2) = 25h^2 \implies h^2 = \tfrac{1600}{9}.$$

So $h = \frac{40}{3}$.

Answer: **E**

Problem 19

Starting with a list of three numbers, form a sequence in which each subsequent term is the average of the previous three terms. For example, starting with the sequence $(2, 4, 9)$, the next two terms are $\frac{2+4+9}{3} = 5$ and $\frac{4+9+5}{3} = 6$. Which of the following initial three terms will produce a sequence that has the greatest 2026th term?

A $(1, 2, 3)$ **B** $(2, 1, 1)$ **C** $(3, 4, 8)$ **D** $(4, 4, 7)$ **E** $(4, 5, 6)$

Solution. For terms x, y, z followed by $w = \frac{x+y+z}{3}$, we have

$$y + 2z + 3w = x + 2y + 3z,$$

so $x + 2y + 3z$ stays the same along the sequence. The recurrence has characteristic equation $3t^3 = t^2 + t + 1$, i.e. $(t-1)(3t^2+2t+1) = 0$, whose other two roots have modulus $\frac{1}{\sqrt{3}}$. So every sequence converges to L with $6L = a + 2b + 3c$, and by the 2026th term the difference from L is astronomically small.

The values of $6L$ are $14, 7, 35, 33, 32$ for A to E. These differ by at least 1, so (C) has the greatest 2026th term.

Answer: **C**

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Problem 20

Which of the following statements most accurately describes the graph of the relationship $\sin(x^2) = \cos(y^2)$?

A the set of all points $\left(\sqrt{\frac{m}{\pi}}, \sqrt{\frac{n+\frac{1}{2}}{\pi}}\right)$, where m and n are nonnegative integers

B the set of all lines $y = \pm x + \frac{\pi}{2} + k\pi$, where k is an integer

C the set of all circles $x^2 + y^2 = \frac{\pi}{2} + 2\pi n$, where n is a nonnegative integer

D a set of infinitely many concentric circles and infinitely many hyperbolas

E a set of finitely many ellipses centered at the origin and infinitely many parabolas

Solution. Write $\sin(x^2) = \cos\left(\frac{\pi}{2} - x^2\right)$. Two angles have equal cosines exactly when they are equal or opposite up to a multiple of 2π, so for some integer k

$$y^2 = \frac{\pi}{2} - x^2 + 2k\pi \quad \text{or} \quad y^2 = x^2 - \frac{\pi}{2} + 2k\pi.$$

The first gives the circles $x^2 + y^2 = \frac{\pi}{2} + 2k\pi$ for every $k \geq 0$, all centred at the origin. The second gives $y^2 - x^2 = 2k\pi - \frac{\pi}{2}$, and since the right side is never 0, each k gives a hyperbola. So the graph is infinitely many concentric circles together with infinitely many hyperbolas.

Answer: D

Problem 21

Let T_n be the nth triangular number, that is, $T_1 = 1$, $T_2 = 1 + 2 = 3$, $T_3 = 1 + 2 + 3 = 6$, and so on. The value of

$$\sum_{k=1}^{\infty} T_k \cdot \left(\frac{1}{3}\right)^k$$

can be expressed as $\frac{p}{q}$, where p and q are relatively prime positive integers. What is $p + q$?

A 14 **B** 15 **C** 16 **D** 17 **E** 18

Solution. Differentiating the geometric series $\sum x^k = \frac{1}{1-x}$ twice gives $\sum \frac{k(k+1)}{2} x^{k-1} = \frac{1}{(1-x)^3}$, so

$$\sum_{k \geq 1} T_k x^k = \frac{x}{(1-x)^3}.$$

At $x = \frac{1}{3}$ this is $\frac{1/3}{8/27} = \frac{9}{8}$, so $p + q = 17$.

Answer: D

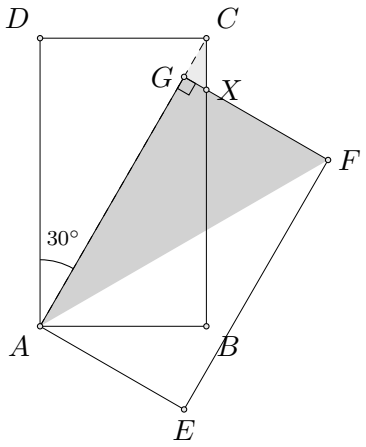
Problem 22

Rectangular card $ABCD$ has $AB = 1$ and $AC = 2$. A second card, $AEFG$, congruent to the first, is stacked onto it so that $\overline{AG}$ lies on $\overline{AC}$ and the two cards share corner A. Additional cards are added, following the same pattern (with shared corner A), creating a fan of cards. The process stops when a new card fits exactly atop $ABCD$. The shape then created is a nonconvex polygon with rotational symmetry. What is the area of this polygon?

A $20\sqrt{3} - 24$ **B** $34\sqrt{3} - 48$ **C** $6 + 3\sqrt{3}$ **D** 12 **E** $8\sqrt{3}$

Solution.

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Since $AB = 1$ and $AC = 2$, we have $AD = BC = \sqrt{3}$ and $\angle CAD = \angle ACB = 30°$. Card $AEFG$ is $ABCD$ rotated 30° about A, so twelve cards bring us back to $ABCD$. The polygon is made of twelve copies of the region $ACXF$, where $X = BC \cap GF$ is the notch between the corners C and F.

Split $ACXF$ along AX. In right triangle GCX we have $GC = 2 - \sqrt{3}$ and $\angle GCX = 30°$, so

$$CX = \frac{2(2-\sqrt{3})}{\sqrt{3}}, \qquad GX = \frac{2-\sqrt{3}}{\sqrt{3}}, \qquad XF = 1 - GX = \frac{2\sqrt{3}-2}{\sqrt{3}}.$$

Triangle ACX has base CX on line BC, at distance $AB = 1$ from A. Triangle AXF has base XF on line GF, at distance $AG = \sqrt{3}$ from A. So

$$[ACXF] = \frac{1}{2} \cdot \frac{2(2-\sqrt{3})}{\sqrt{3}} + \frac{1}{2} \cdot \sqrt{3} \cdot \frac{2\sqrt{3}-2}{\sqrt{3}} = \frac{2\sqrt{3}-3}{3} + \sqrt{3} - 1 = \frac{5\sqrt{3}-6}{3}.$$

The area of the polygon is $12 \cdot \frac{5\sqrt{3}-6}{3} = 20\sqrt{3} - 24$.

Answer: A

Problem 23

For positive integers a and b, let $R(a, b)$ denote the remainder when a is divided by b. Suppose N is the least positive integer such that

$$R(N, 2) \geq 1, \quad R(N, 4) \geq 2, \quad R(N, 8) \geq 4, \quad \ldots, \quad R\left(N, 2^{200}\right) \geq 2^{199}.$$

What are the last three digits of N?

A 125 **B** 249 **C** 375 **D** 511 **E** 875

Solution. $R(N, 2^k)$ is the number formed by the last k binary digits of N, and it is at least 2^{k-1} exactly when the kth digit from the right is 1. So all of the last 200 binary digits of N are 1, and the least such N is

$$N = 2^{200} - 1.$$

Now find 2^{200} mod 1000 by repeated squaring:

$$2^{10} \equiv 24, \quad 2^{20} \equiv 576, \quad 2^{40} \equiv 776, \quad 2^{80} \equiv 176, \quad 2^{160} \equiv 976.$$

So $2^{200} = 2^{160} \cdot 2^{40} \equiv 976 \cdot 776 \equiv 376$, and N ends in 375.

Answer: C

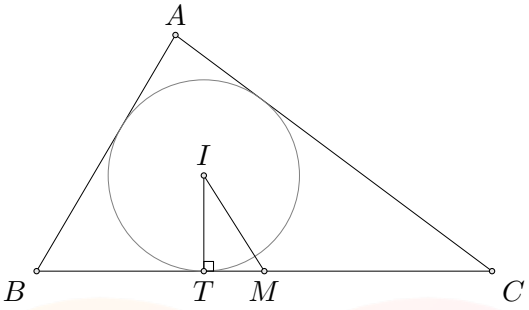
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Problem 24

Let $\triangle ABC$ be a scalene triangle with incenter I. The distances from I to the midpoints of the sides of $\triangle ABC$ are $\sqrt{11}$, $\sqrt{14}$, and $\sqrt{19}$. What is the sum of the possible areas of the circle inscribed in $\triangle ABC$?

A 10π **B** $\frac{44\pi}{3}$ **C** $\frac{58\pi}{3}$ **D** $\frac{88\pi}{3}$ **E** 30π

Solution.



Let the tangent lengths from A, B, C be z, x, y. On BC the touch point is x from B and the midpoint is $\frac{x+y}{2}$ from B, so they are $\frac{|x-y|}{2}$ apart. By Pythagoras, the distance from I to that midpoint satisfies

$$(\text{distance})^2 = r^2 + \tfrac{1}{4}(x-y)^2,$$

and likewise for the other two sides.

Of $|x-y|$, $|y-z|$, $|z-x|$, the largest is the sum of the other two, and it goes with the largest distance $\sqrt{19}$. So

$$\sqrt{19-r^2} = \sqrt{11-r^2} + \sqrt{14-r^2}.$$

Squaring twice gives $3r^4 - 88r^2 + 580 = 0$, so $r^2 = 10$ or $r^2 = \frac{58}{3}$. The second is impossible since $r^2 < 11$.

A triangle with $r^2 = 10$ does exist. The half-differences are then $1, 2, 3$, so take $x = z+2$ and $y = z+6$, and we need $\frac{xyz}{x+y+z} = r^2 = 10$, which is $z^3 + 8z^2 - 18z - 80 = 0$. This cubic changes sign between $z = 3$ and $z = 4$, so a positive root exists. The answer is 10π.

Answer: A

Problem 25

A *finite integer tower* is an expression of the form $a_1^{a_2^{\cdot^{\cdot^{a_n}}}}$, where $n \geq 1$ and $a_1, a_2, \ldots, a_n$ are positive integers greater than 1. (Exponentiation is performed from the top down. For example, 3^{2^3} equals 3^8, not 9^3.) Two towers are considered different if they have a different number of layers or at least one pair of numbers in corresponding positions differ. Define $f(x)$ to be the number of different finite integer towers whose value is x. (For example, $f(16) = 4$ because 4 towers evaluate to 16, namely 16, 2^4, 4^2, and 2^{2^2}.) What is $f\left(2^{8192}\right)$?

A 17 **B** 32 **C** 42 **D** 43 **E** 47

Solution. Peel off the bottom layer. A tower worth x is either just x, or a raised to a tower worth k, where $a^k = x$ and $a, k \geq 2$. So

$$f(x) = 1 + \sum_{a^k = x,\ k \geq 2} f(k),$$

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and for powers of two this reads $f(2^j) = 1 + \sum_{k|j,\ k\geq 2} f(k)$.

We only need $f(2^j)$ for $j \leq 13$, and the only perfect powers up to 13 are $4, 8, 9$, each with $f = 2$; every other k has $f(k) = 1$. Running through the divisors:

j	1	2	3	4	5	6	7	8	9	10	11	12	13
$f(2^j)$	1	2	2	4	2	4	2	6	4	4	2	7	2

Finally $2^{8192} = a^k$ with $k \geq 2$ forces $k = 2^i$ for $1 \leq i \leq 13$, so

$$f\left(2^{8192}\right) = 1 + \sum_{i=1}^{13} f(2^i) = 1 + 42 = 43.$$

Answer: **D**

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